

LAB REPORT: FOOTPRINTING & RECONNAISSANCE ATTACKS

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<https://github.com/Diksha0665/Networkwalks-B082-week2--Cybersecurity-Footprinting-and-Network-Scanning/tree/main>

- Batch: B082
- Module: W2-PM1
- Target: networkwalks.com
- Date: 20 August 2026

1. Letter of Authorization & Legal Scope

Reference No. & Date	NW-LOA-B082-017 Date of Issue: 17 August 2026
Client / Target Owner	Networkwalks (Target Domain: networkwalks.com & Authoritative DNS)
Authorized Scope & Tools	Passive & Active Reconnaissance: whois, whatweb, nslookup, curl, wafw00f, dnsrecon, non-destructive subnet mapping.
Strictly Prohibited Activities	Exploitation, unauthorized access, privilege escalation, Denial-of-Service (DoS), data alteration/theft, or social engineering.

2. Legal & Ethical Compliance Notice

All reconnaissance and scanning activities documented in this report were conducted strictly in compliance with the Letter of Authorization (Ref: NW-LOA-B082-017) issued by Networkwalks. Testing remained strictly within the authorized boundaries of passive intelligence gathering and non-destructive service fingerprinting. Unauthorized reconnaissance against third-party systems without prior written consent is strictly prohibited under international cybercrime legislation.

<https://drive.google.com/drive/folders/1UEGrpEvAmFxf6HUI6arn6kXWkP-Hb5D>

3. Liability Disclaimer

These materials and commands are documented exclusively for educational, technical training, and authorized security auditing purposes. The author (Diksha), Networkwalks, and affiliated instructors assume no responsibility or liability for any misuse, operational damage, or legal consequences resulting from the execution of the methodologies, commands, or data contained herein.

4. Executive Overview

Reconnaissance (footprinting) is the foundational phase of any ethical hacking assessment. Before interacting directly with target infrastructure, publicly available and direct non-destructive telemetry is collected to systematically map the target's attack surface. This engagement profiles **networkwalks.com** using native Kali Linux reconnaissance utilities to extract registrar metadata, IP allocations, web software frameworks, header properties, active firewalls, and DNS architecture.

5. Tools Matrix

Tool	Focus Area	Mode	Utility & Purpose
whois	Registry Intelligence	Passive	Query registrar information, registration/expiration dates, and authoritative name servers.
whatweb	Web Stack Profiling	Passive / Active	Fingerprint web server software, CMS version, plugins, JavaScript libraries, and public IP.
nslookup	DNS Querying	Passive	Resolve hostnames directly to target IPv4 addresses via recursive resolvers.
curl	Header Analysis	Passive	Inspect raw HTTP/HTTPS response headers to identify server banners, caching engines, and cookies.
wafw00f	Firewall Detection	Active Probe	Analyze behavioral HTTP response signatures to detect active Web Application Firewalls (WAF).
dnsrecon	DNS Enumeration	Active Enumeration	Enumerate DNS records including MX, A, TXT, SPF policies, and autodiscovery SRV services.

Practical Lab Execution & Screenshot Evidence

Task 1: Domain Registration Intelligence (whois)

- **Objective:** Query public domain registration records to identify the registrar, creation/expiration dates, and primary name servers.

```
$ whois networkwalks.com
```

```

(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr
ohibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh
ibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr
ohibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
cf/
>>> Last update of whois database: 2026-08-19T16:53:14Z <<<

For more information on Whois status codes, please visit https://icann.org/ep
p

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiratio
n
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only

```

Extracted Technical Findings:

- **Registrar:** GoDaddy.com, LLC (IANA ID: 146)
- **Creation Date:** 2019-11-06T22:51:46Z
- **Updated Date:** 2025-11-12T10:08:43Z
- **Registry Expiry Date:** 2027-11-06T22:51:46Z
- **Name Servers:** NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM
- **Hosting Infrastructure:** HostGator infrastructure confirmed via authoritative DNS delegations.

Task 2: Web Technology Stack Fingerprinting (whatweb)

- **Objective:** Identify the underlying web server, CMS, plugins, client-side libraries, and public IP.

```
$ whatweb networkwalks.com
```

```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]
```

Extracted Technical Findings:

- **Target IP Address:** 192.232.216.135
- **Web Server Daemon:** Apache (HTTP 200 / 301 Redirect to HTTPS)
- **CMS & Version:** WordPress 7.0.4
- **Installed Plugins:** WordPress Download Manager 3.3.58
- **Client Frameworks:** Bootstrap 7.0.4, JQuery 3.7.1, HTML5
- **Exposed Contact Information:** info@networkwalks.com

Task 3: DNS Host Resolution (nslookup)

- **Objective:** Query DNS servers to resolve the domain to its corresponding IP address.

```
$ nslookup networkwalks.com
```

```
(kali@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Extracted Technical Findings:

- **DNS Server Queried:** 8.8.8.8#53 (Google Public DNS)
- **Resolved Public IPv4:** 192.232.216.135
- **Answer Type:** Non-authoritative answer

Task 4: HTTP Response Headers & Server Inspection (curl)

- **Objective:** Read raw HTTP response headers to identify server banners, security flags, caching layers, and API endpoints.

```
$ curl -I https://networkwalks.com
```

```
(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=3a4d56add0fcfa2a4cdd5ab42a5463bd; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 19 Aug 2026 17:17:51 GMT
server: Apache
```

Extracted Technical Findings:

- **Protocol & Status Code:** HTTP/2 200
- **Web Server Banner:** Apache
- **Caching Architecture:** x-nginx-cache: WordPress, x-endurance-cache-level: 0
- **API Disclosure:** Link header exposes WordPress REST API at <https://networkwalks.com/wp-json/>
- **Cookie Configuration:** Set-Cookie: wpdm_client; path=/; domain=networkwalks.com; secure; HttpOnly

Task 5: Web Application Firewall Detection (wafw00f)

- **Objective:** Determine if the target web server is protected by a Web Application Firewall (WAF).

```
$ wafw00f networkwalks.com
```

```
(kali㉿kali)-[~]  
$ wafw00f networkwalks.com
```



```
404 Hack Not Found  
405 Not Allowed  
403 Forbidden  
502 Bad Gateway 500 Internal Error
```

```
~ WAFW00F : v2.4.2 ~  
The Web Application Firewall Fingerprinting Toolkit
```

```
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2
```

```
(kali㉿kali)-[~]  
$
```

Extracted Technical Findings:

- **WAF Status:** Active Web Application Firewall detected

- **Firewall Identity:** ModSecurity (SpiderLabs) WAF
- **Security Implication:** Active payload inspection and attack filtering is enforced by ModSecurity rule engines.

Task 6: Comprehensive DNS Enumeration (dnsrecon)

- **Objective:** Enumerate domain name servers, mail records, sender authentication policies, and service records.

```
$ dnsrecon -d networkwalks.com
```

```
(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-08-21T02:53:02.509256-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-21T02:53:02.509832-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-21T02:53:03.124326-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-21T02:53:03.463596-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-21T02:53:03.772343-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-21T02:53:04.311530-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-21T02:53:04.311765-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-21T02:53:04.850257-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-21T02:53:05.598497-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-21T02:53:06.096863-0400 INFO A networkwalks.com 192.232.216.135
2026-08-21T02:53:07.133414-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-21T02:53:07.133573-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-21T02:53:07.759844-0400 INFO Enumerating SRV Records
2026-08-21T02:53:10.793805-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-21T02:53:10.794255-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-21T02:53:10.794321-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-21T02:53:10.794359-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-21T02:53:10.794390-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-21T02:53:10.794420-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-21T02:53:10.794447-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-21T02:53:10.794472-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-21T02:53:10.795812-0400 INFO 8 Records Found
2026-08-21T02:53:10.796045-0400 INFO Completed enumeration for domain: networkwalks.com
```

Extracted Technical Findings:

- **A Record:** networkwalks.com -> 192.232.216.135
- **MX (Mail Exchange):** mail.networkwalks.com -> 192.232.216.135
- **Nameserver Software:** BIND Version 9.16.23-RH identified on 50.87.144.87 and 192.232.216.131
- **SPF Policy:** v=spf1 a mx ip4:50.87.144.87 include:websitewelcome.com ~all
- **Discovered Services:** 8 cPanel email autodiscovery SRV records identified (cpanelmaildiscovery.cpanel.net).

6. Comprehensive Target Profile Summary

Intelligence Category	Discovered Profile Detail
Domain & Resolved Host IP	networkwalks.com / 192.232.216.135
Registrar	GoDaddy.com, LLC (IANA ID: 146)
Hosting Infrastructure	HostGator / cPanel Shared Hosting Architecture
Web Server Daemon	Apache with Nginx caching reverse-proxy layer
CMS Platform & Plugins	WordPress 7.0.4 with WP Download Manager 3.3.58
API Endpoints Disclosed	WordPress REST API exposed at /wp-json/
DNS Server Software	BIND 9.16.23-RH (HostGator Name Servers)
Perimeter Defense	ModSecurity (SpiderLabs) Web Application Firewall active

7. Defensive Remediation & Hardening Recommendations

- **Information Leakage Suppression:** Suppress web server banner tokens (Server: Apache) and custom caching headers (x-nginx-cache, x-endurance-cache-level) across HTTP responses to deny attackers passive technology identification.
- **Restrict Public REST API Enumeration:** Protect or restrict sensitive endpoints under `/wp-json/` (e.g., `/wp-json/wp/v2/users`) using authentication gateways to prevent unauthorized user and plugin discovery.

- **DNS Version Masking:** Configure the BIND DNS server configuration (``named.conf``) with ``version none;`` or a custom string to suppress exact BIND version disclosure (9.16.23-RH).
- **WAF Rule Maintenance:** Continuously tune and update ModSecurity WAF rule sets (such as the OWASP ModSecurity Core Rule Set) to prevent signature evasion and zero-day bypass techniques.